Automatic Street Light Control System using LDR and Arduino (Tinkercad Simulation)

Abstract

The aim of this project is to design and simulate an automatic street light control system using a Light Dependent Resistor (LDR) and Arduino in Tinkercad. The system automatically turns ON the light during low light conditions (night) and turns OFF during high light conditions (day), thereby conserving energy. This simulation demonstrates the basic concept of IoT-based smart lighting systems.

Objective

- To simulate a sensor-based IoT system using Tinkercad
- To control an LED using a light sensor (LDR)
- To understand automatic lighting systems

Components Used

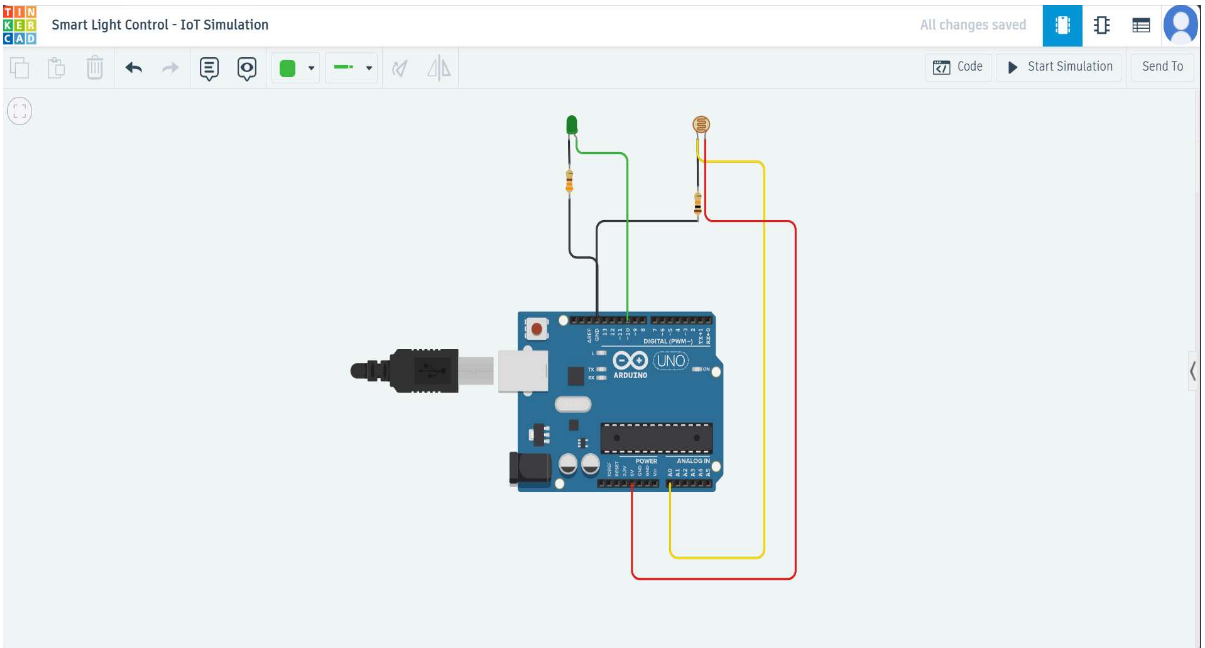
- Arduino Uno
- LDR (Light Dependent Resistor)
- LED
- Resistor (330Ω for LED)
- Resistor (10kΩ for LDR)
- Jumper wires

Working Principle

The system works based on light intensity detection. The LDR senses the surrounding light and provides an analog signal to the Arduino. The Arduino processes this signal and decides whether to turn the LED ON or OFF. When the environment is dark, the LED turns ON, and when the environment is bright, the LED turns OFF. This helps in automatic and energy-efficient lighting control.

Circuit Description

The LDR and a 10kΩ resistor form a voltage divider circuit. The output of this circuit is connected to analog pin A0 of the Arduino. The LED is connected to digital pin 10 through a 330Ω resistor. The Arduino reads the analog value from pin A0 and controls the LED connected to pin 10. All components are directly connected using jumper wires in the simulation environment.

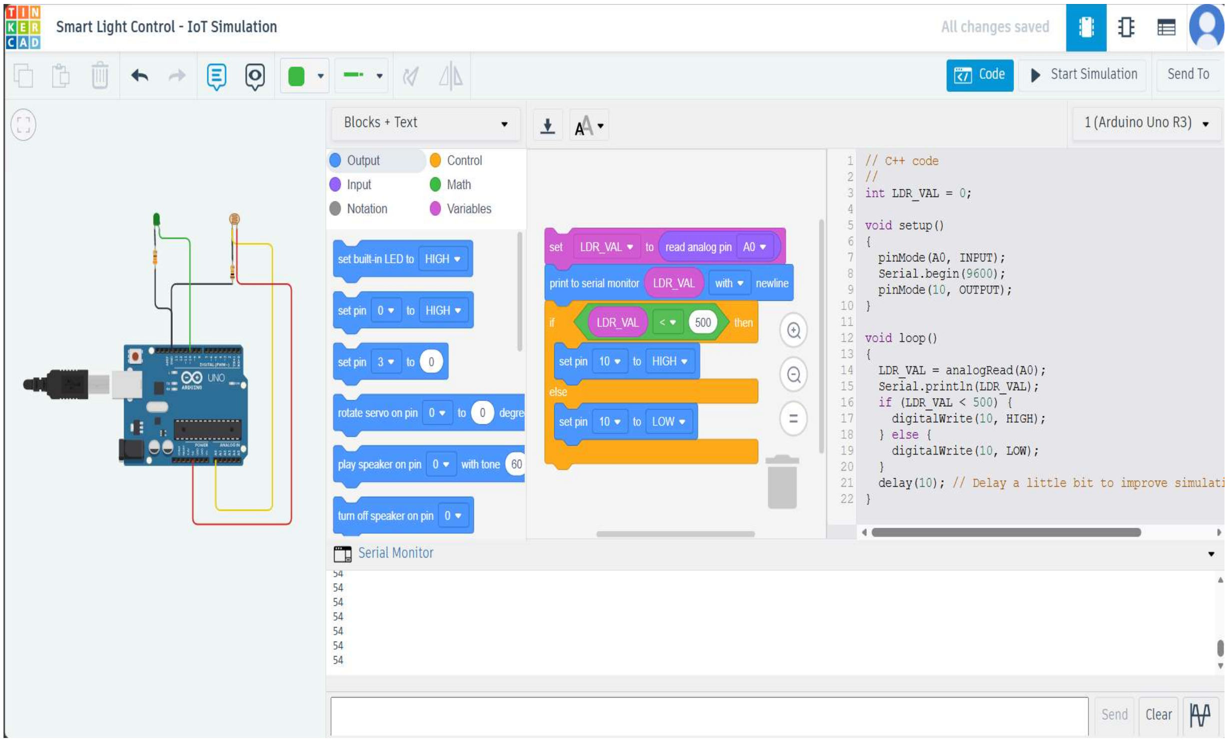


Code

```
// C++ code
//
int LDR_VAL = 0;

void setup()
{
  pinMode(A0, INPUT);
  Serial.begin(9600);
  pinMode(10, OUTPUT);
}

void loop()
{
  LDR_VAL = analogRead(A0);
  Serial.println(LDR_VAL);
  if (LDR_VAL < 500) {
    digitalWrite(10, HIGH);
  } else {
    digitalWrite(10, LOW);
  }
  delay(10); // Delay a little bit to improve simulation performance
}
```



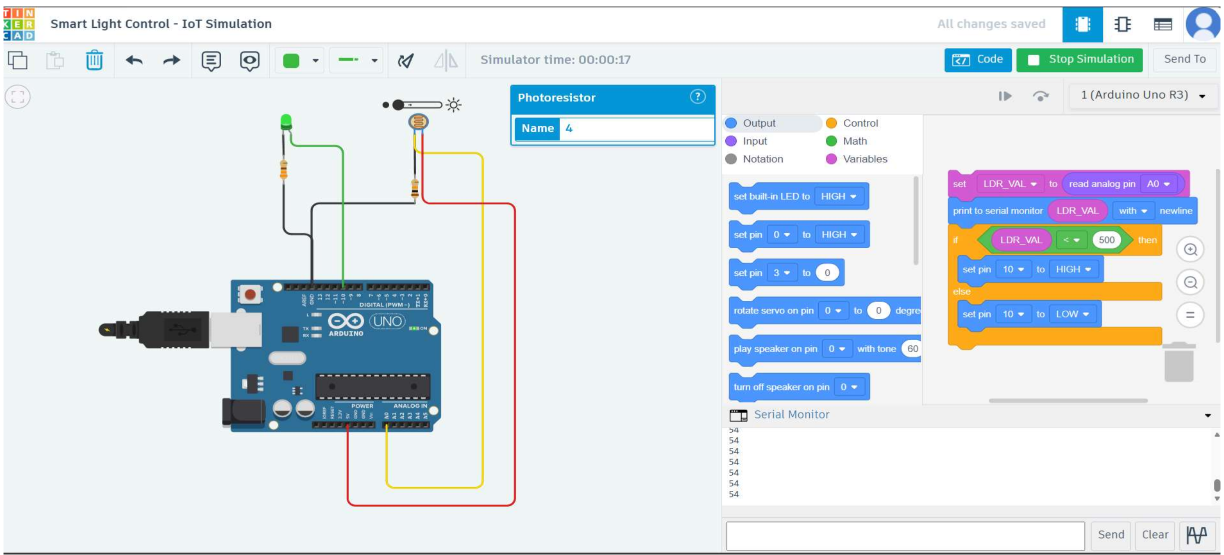
Code Explanation

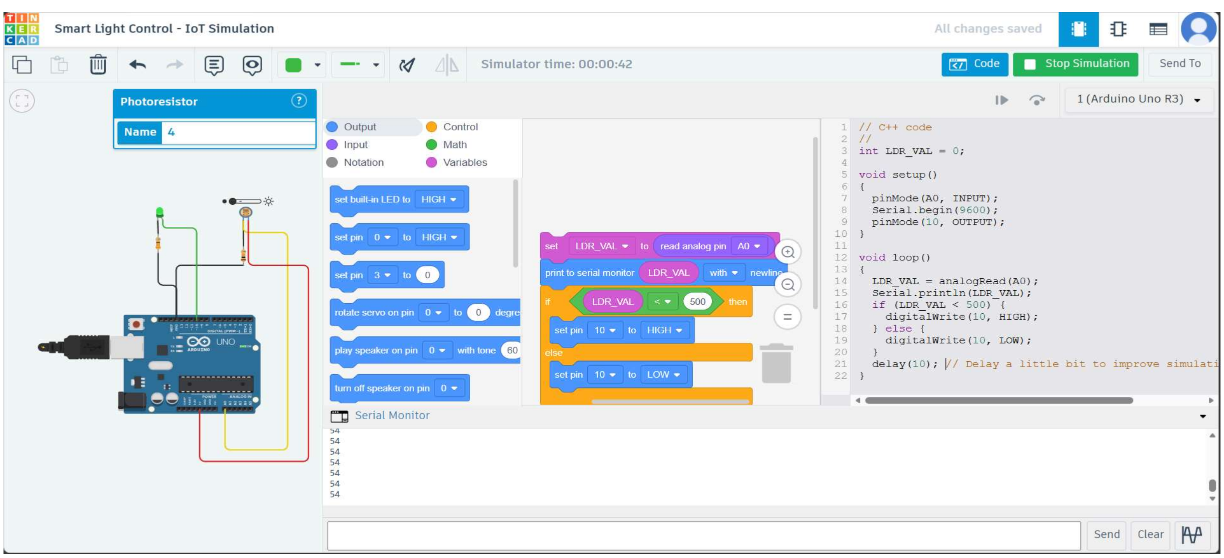
- `LDR_VAL` is a variable used to store the light intensity value
- `analogRead(A0)` reads the analog value from the LDR sensor (range 0–1023)
- Analog pin A0 is used to read sensor input
- The `if` condition checks whether the environment is dark or bright
- If the value is less than 500 (dark), the LED is turned ON
- If the value is greater than 500 (bright), the LED is turned OFF
- `digitalWrite()` controls the LED
- Digital pin 10 is used to control the LED
- `Serial.println()` displays the sensor values in the serial monitor

The code continuously reads the light intensity from the LDR sensor and processes the data using conditional statements. Based on the input, the Arduino automatically controls the LED, ensuring that the light turns ON in darkness and OFF in brightness. This demonstrates the working of an automated lighting system.

Observation

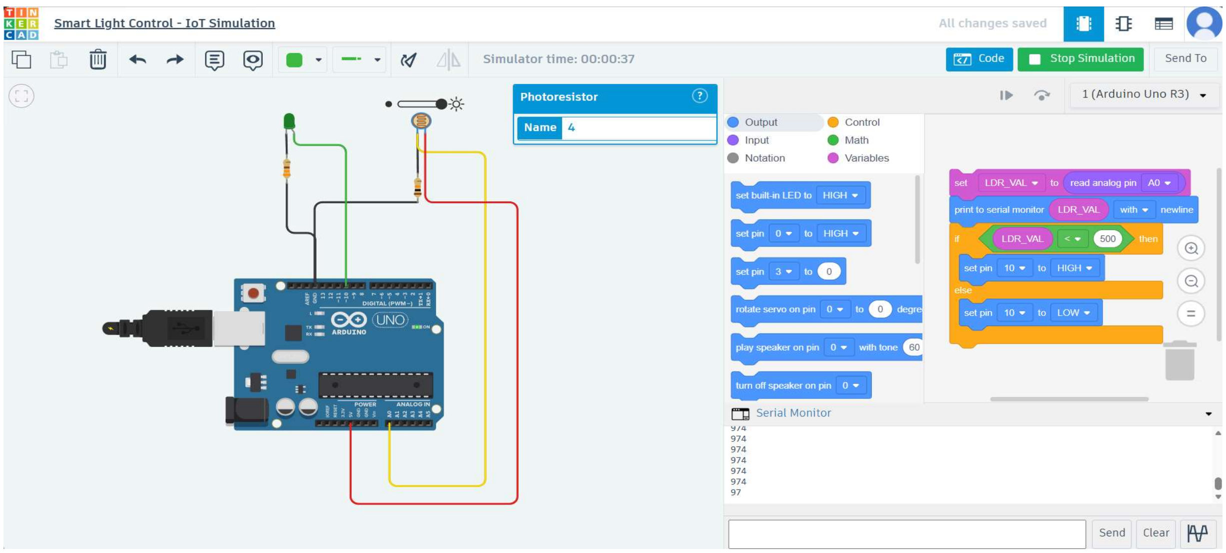
- In dark condition → LDR value ≈ **54** → LED **ON**

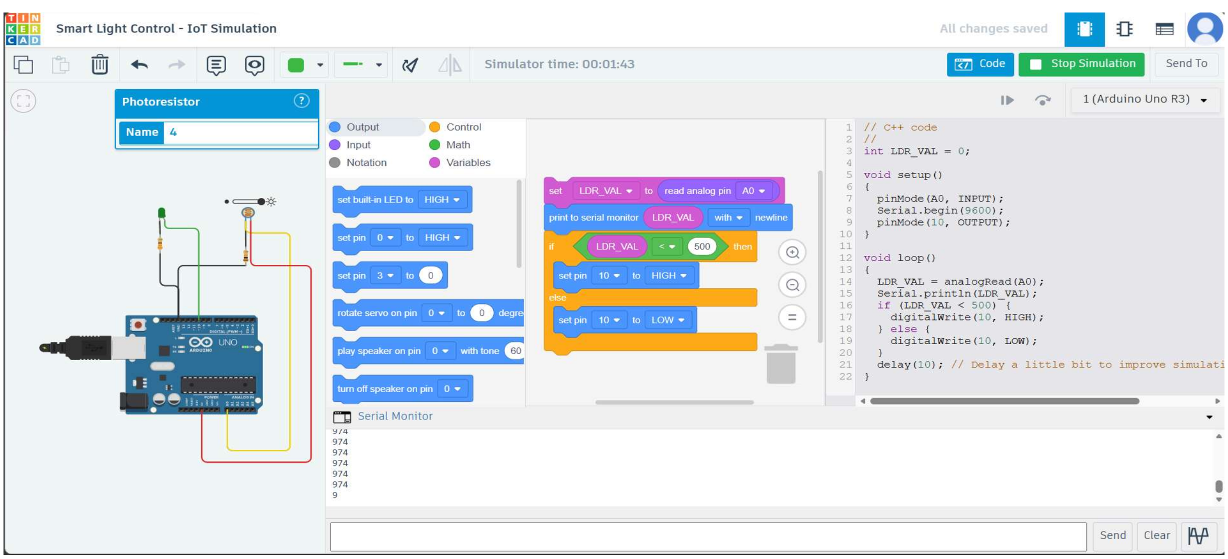




When the light intensity is low, the LDR provides a lower analog value (approximately 54). Since this value is less than the threshold (500), the Arduino turns the LED ON. This represents night-time operation.

- In bright condition → LDR value ≈ **974** → LED **OFF**





When the light intensity is high, the LDR provides a higher analog value (approximately 974). Since this value is greater than the threshold (500), the Arduino turns the LED OFF. This represents daytime operation.

Applications

- Automatic street lighting systems
- Smart home lighting
- Energy-saving automation systems

Conclusion

The project successfully demonstrates an automatic street light control system using an LDR sensor and Arduino. It effectively controls lighting based on environmental conditions, reducing energy consumption and human effort.

Future Scope

This system can be further enhanced using IoT platforms such as Blynk or ThingSpeak for remote monitoring and control.

Simulation Link

The complete simulation can be accessed using the following link:

https://www.tinkercad.com/things/hbKosF69WKQ/editel?sharecode=FnL50j_E3c_5xm_8yIijxQacjLm0TVUtZ4b6yit9hiE